

Ratio of areas of triangle

$$\frac{A_1}{A_2} = \frac{b_1 \times h_1}{b_2 \times h_2}$$

If base are same $\frac{A_1}{A_2} = \frac{h_1}{h_2}$

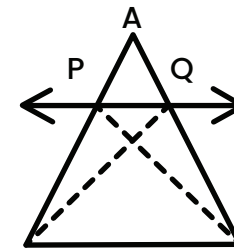
If height are same $\frac{A_1}{A_2} = \frac{b_1}{b_2}$

• If both are same $A_1 = A_2$

Basic proportionality theorem (BPT)

If $PQ \parallel BC$, then

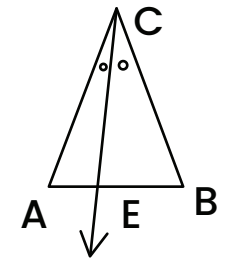
$$\frac{AP}{PB} = \frac{AQ}{QC}$$



Property of an angle bisector of triangle

If CE is angle bisector of $\angle C$ on side AB then

$$\frac{AE}{EB} = \frac{CA}{CB}$$



Similarity

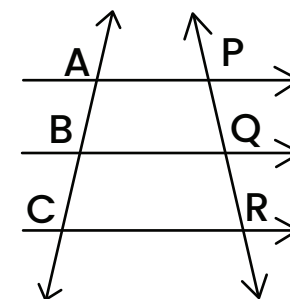
Test of similarity

- AAA test (If corresponding angles are equal)
- AA test (If two angles are equal)
- SSS test (If corresponding sides are proportion)
- SAS test (If included angle is equal and true sides are in same proportion)

Property of three parallel lines

If line $l \parallel$ line $m \parallel$ line n then

$$\frac{AB}{BC} = \frac{PQ}{QR}$$



Therem of area of similar triangle

If $\triangle ABC \sim \triangle PQR$ then

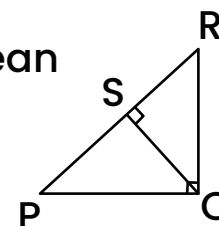
$$\frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{AC^2}{PR^2}$$

Pythagoras Triplet

If a, b, c are natural number and $a > b$ then $[(a^2+b^2), (a^2-b^2), 2ab]$ is Pythagoras triplate

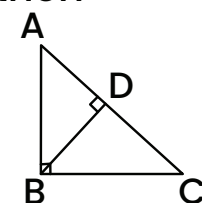
Theorem of geometric mean

$QS^2 = PS \times SR$
i.e. QS is geometric mean of seg PS and SR



Similarity in right angled triangle

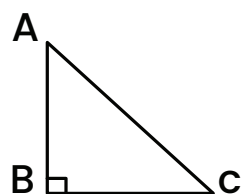
If $\angle ABC = 90^\circ$ & $BD \perp AC$ then
1. $\triangle ADB \sim \triangle ABC$
2. $\triangle BDC \sim \triangle ABC$
3. $\triangle ADB \sim \triangle BDC$



Pythagoras Theorem

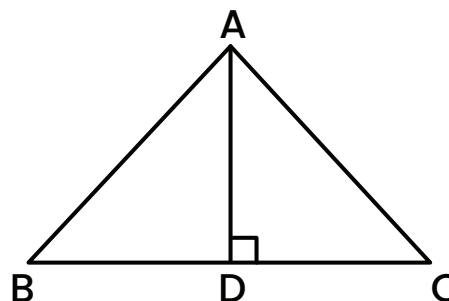
Pythagoras theorem

In a right angled triangle the square of hypotenuse is equal to sum of the squares of remaining true sides.
i.e. $AC^2 = AB^2 + BC^2$



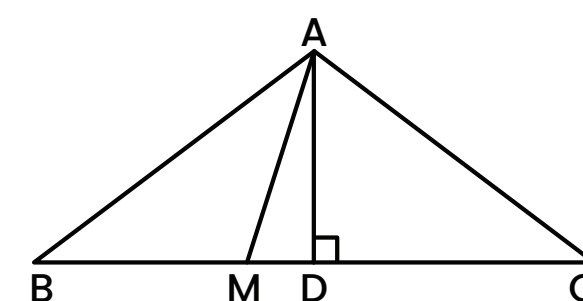
Application of Pythagoras theorem

$$AB^2 = BC^2 + AC^2 - 2BC \times DC$$



Apollonius theorem

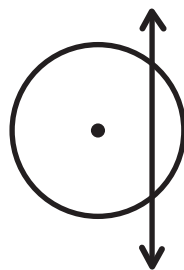
$$AB^2 + AC^2 = 2AM^2 + 2BM^2$$



Circle passing through

- One point – Infinite
- Two points – Infinite
- Three points – Only one
- No circle can pass through 3 collinear points

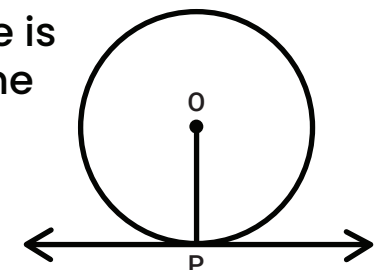
Secant



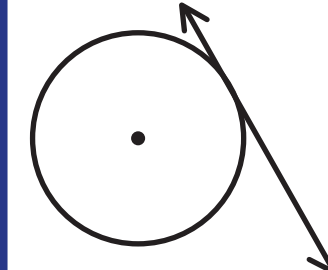
Tangent Theorem

A tangent at any point of a circle is perpendicular to the radius at the point of contact.

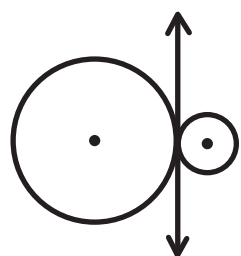
i.e. $OP \perp \text{line } \ell$



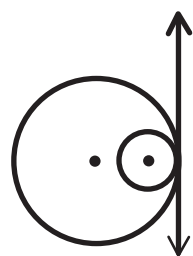
Tangent



Touching Circles



Touching externally

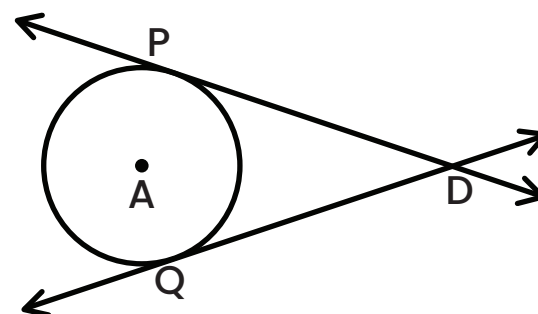


Touching internally

Theorem

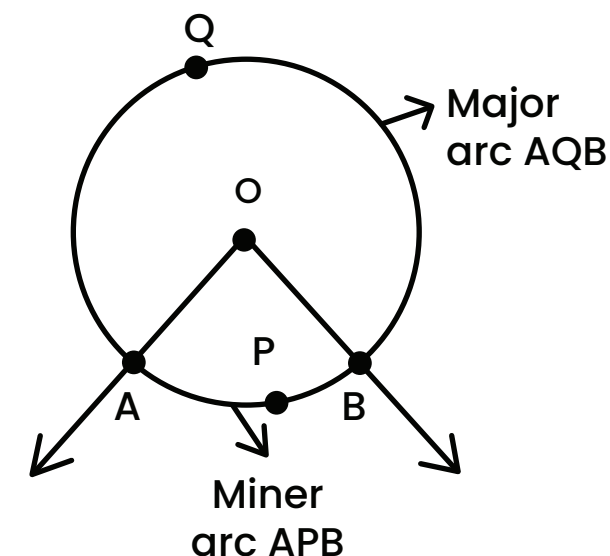
If two circles touch each other, their point of contact lies on the line joining their centres.

Tangent segment Theorem

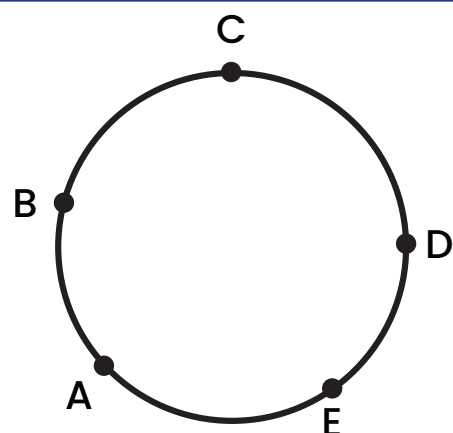


Tangent segment drawn from an external point to a circle are congruent
i.e. $\text{seg } DP \cong \text{seg } DQ$.

Arc of a circle



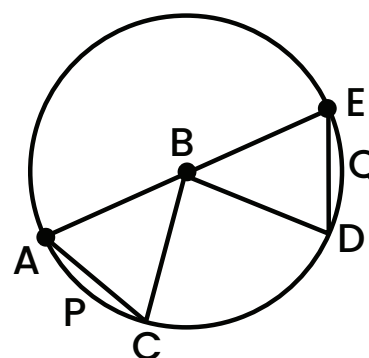
Property of sum of measure of arcs



$$m(\text{arc } ABC) + m(\text{arc } CDE) = m(\text{arc } ACE)$$

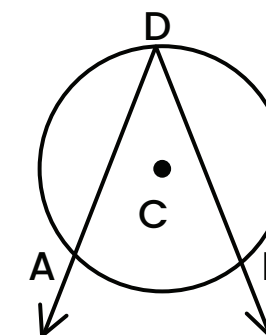
Theorem

The chords corresponding to congruent arcs of circle are congruent.



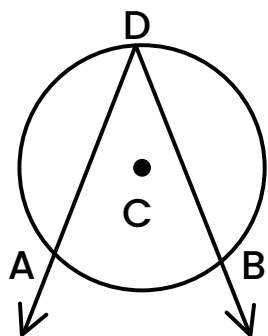
Inscribed angle

$\angle ADB$ is inscribed in the arc ADB.



Inscribed angle theorem

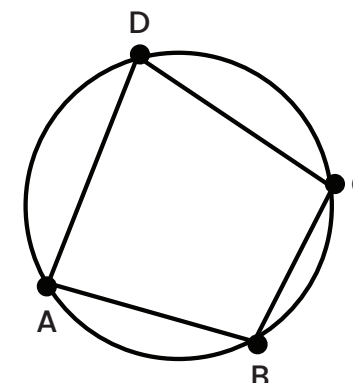
$$\angle BAC = \frac{1}{2} m(\text{arc BDC})$$



Corollaries of inscribed angle theorem

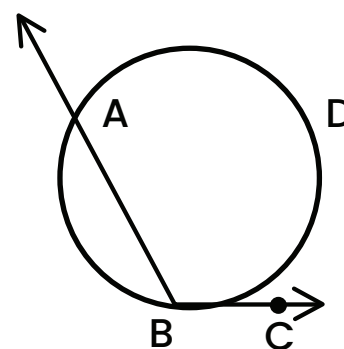
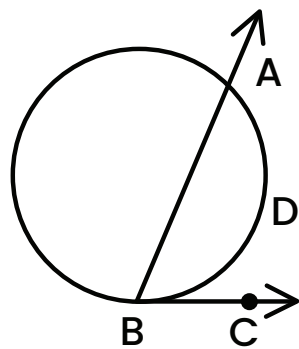
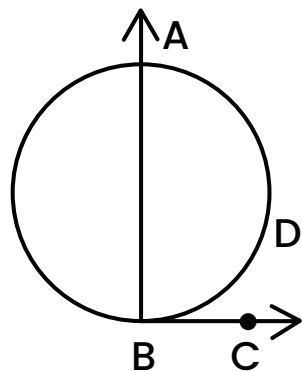
- Angles inscribed in same arc are congruent.
- Angle inscribed in semicircle is a right angle.

Cyclic Quadrilateral



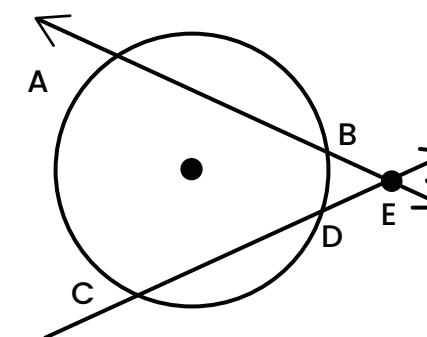
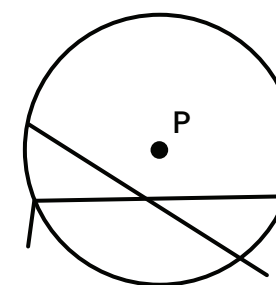
$$\angle B + \angle D = \angle A + \angle C = 180^\circ$$

Theorem of angle between tangent and secant



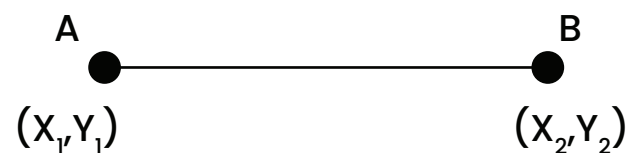
$$\angle ABC = \frac{1}{2} m(\text{arc ADB})$$

Theorem of angle between tangent and secant



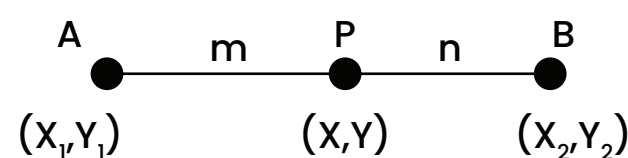
$$AE \times EB = CE \times ED$$

Distance Formula



$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

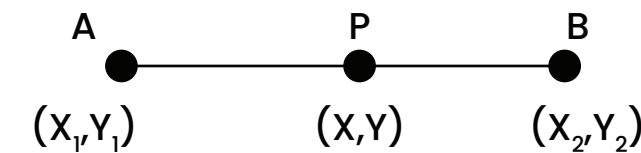
Section Formula



$$\therefore x = \frac{mx_2 + nx_1}{m+n}$$

$$y = \frac{my_2 + ny_1}{m+n}$$

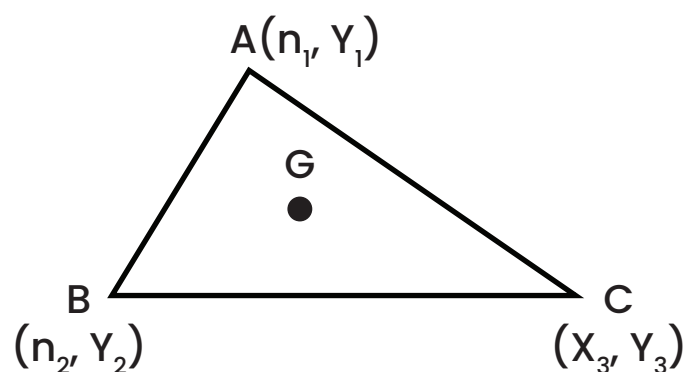
Midpoint Formula



$$\Rightarrow P \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

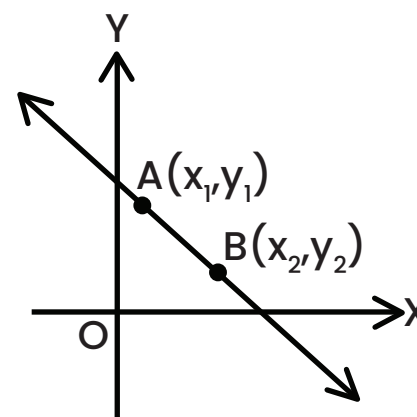
CO-ORDINATE GEOMETRY

Midpoint Formula



$$\Rightarrow G \equiv \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

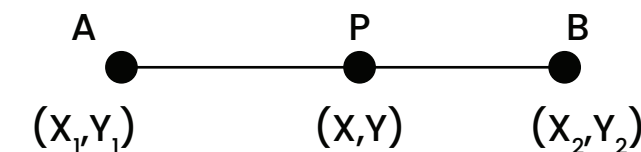
Slope of line



Slope of AB is

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Midpoint Formula



$$\Rightarrow P \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Trigonometry

Trigonometric Ratio

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypo}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypo}}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\text{Opposite}}{\text{Adjacent}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}, \sec \theta = \frac{1}{\cos \theta}$$

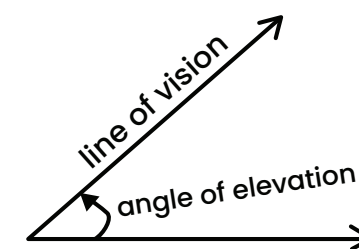
$$\cot \theta = \frac{1}{\tan \theta}$$

Trigonometric Identities

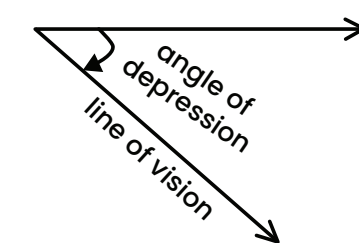
- $\sin^2 \theta + \cos^2 \theta = 1$
- $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
- $1 + \tan^2 \theta = \sec^2 \theta$

Application of Trigonometry

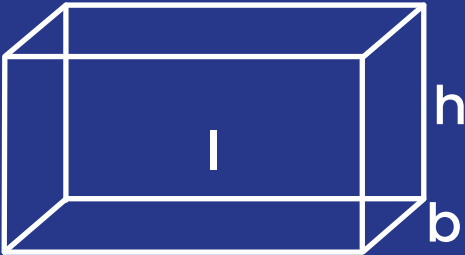
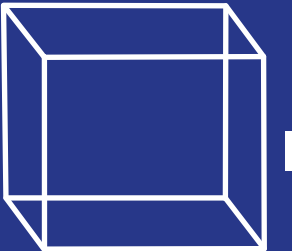
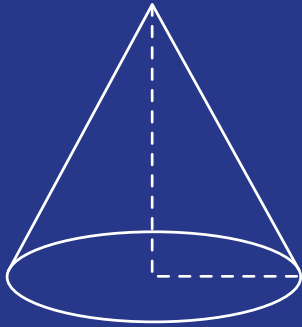
Angle of Elevation

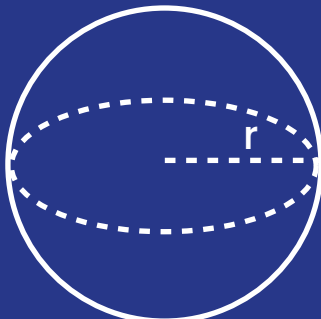
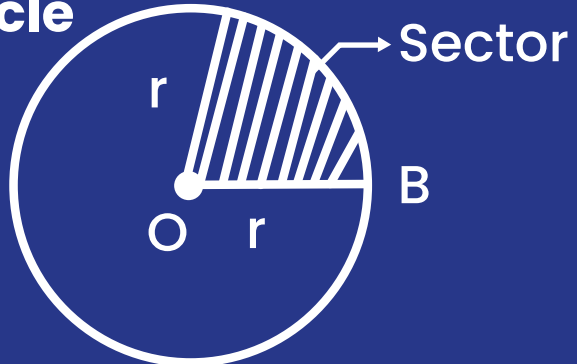


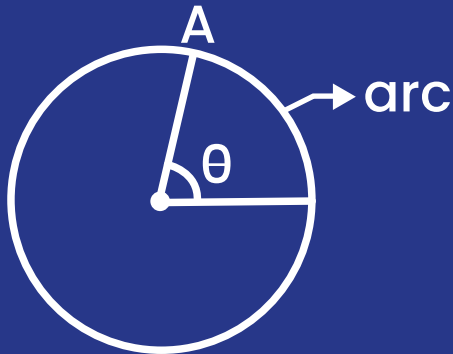
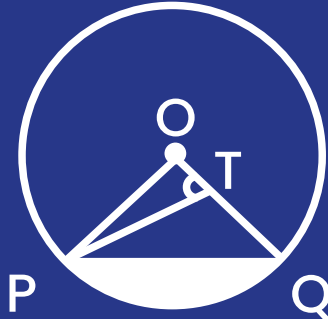
Angle of Depression



Trigonometric Ratios	Angle (θ)				
	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$\operatorname{cosec} \theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\cot \theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

No	Three Dimensional figure	Formulae
1.	Cuboid 	Lateral surface area = $2h(l + b)$ Total surface area = $2(lb + bh + hl)$ Volume = lhb
2.	Cube 	Lateral surface area = $4l^2$ Total surface area = $6l^2$ Volume = l^3
3.	Cylinder 	Curved surface area = $2\pi rh$ Total surface area = $2\pi r(r + h)$ Volume = $\pi r^2 h$
4.	Cone 	Slant height (l) = $\sqrt{h^2 + r^2}$ Curved surface area = πrl Total surface area = $\pi r(r + l)$ Volume = $\frac{1}{3} \times \pi r^2 h$

No	Three Dimensional figure	Formulae
5.	Sphere 	Surface area = $4\pi r^2$ Volume = $\frac{4}{3}\pi r^3$
6.	Hemisphere 	Curved surface area = $2\pi r^2$ Total surface area of solid hemisphere = $3\pi r^2$
7.	Frustum 	Slant height = $l = \sqrt{h^2 + (r_1 - r_2)^2}$ Curved surface area = $\pi l (r_1 + r_2)$ Total surface area = $= \pi l (r_1 + r_2) + \pi r_1^2 + \pi r_2^2$ Volume of frustum $= \frac{1}{3}\pi h (r_1^2 + r_2^2 + r_1 \times r_2)$
8.	Sector of Circle 	Area of sector = $\frac{\theta}{360^\circ} \times \pi r^2$

No	Three Dimensional figure	Formulae
9.	Length of arc 	Length of arc $l = \frac{\theta}{360^\circ} \times 2\pi r$
10.	Length of arc 	Area of segment $P \times Q$ $= r^2 \left[\frac{\pi\theta}{360} - \frac{\sin\theta}{2} \right]$